DO NOW · AO1 REVISION

Density of materials

Particle model of matter
Paper 1 · Combined and Separate
13 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Define density.**	The mass per unit volume of a substance.	☐
2	**State the standard unit of volume.**	Cubic metre (m^3)	☐
3 ★	**State the standard unit of density.**	Kilogram per cubic metre (kg/m^3)	☐
4	**Which state of matter can be compressed easily?**	Gas	☐
5	**Explain why solids and liquids cannot be compressed easily.**	Their particles are already touching.	☐
6 ★	**Describe the motion of particles in a solid.**	They vibrate around a fixed position.	☐
7	**Describe the motion of particles in a liquid.**	They move randomly, sliding past one another.	☐
8	**Describe the motion of particles in a gas.**	They move quickly in random directions.	☐
9	**State the equation linking density, mass and volume.**	density = mass ÷ volume ($\rho = m \div V$)	☐
10	**Describe the arrangement of the particles in a solid.**	Very close together in a regular pattern.	☐
11	**Describe the arrangement of the particles in a liquid.**	Close together, in a random arrangement.	☐
12	**Describe the arrangement of the particles in a gas.**	Far apart, in a random arrangement.	☐
13 ★	**Explain why gases have a much lower density than solids and liquids.**	The particles are far apart, so there is less mass in a given volume.	☐

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